

LAB-9: Data Analytics using Neo4j

Objective: Students will learn how to structure data in a graph format, use Cypher queries for data manipulation, and explore real-world use cases of graph databases.

The objective of this assignment is to give students practical exposure to graph databases by building a social network model using Neo4j.

Assignment:

Design the database for Social networks.

1. Nodes:

- User Node: Represents the users of the social network.
 - Properties: `user_id`, `name`, `age`, `location`, `email`
- Post Node: Represents a post created by a user.
 - Properties: `post_id`, `content`, `timestamp`, `likes_count`, `comments_count`
- Comment Node: Represents a comment on a post.
 - Properties: `comment_id`, `content`, `timestamp`
- Location Node: (Optional) If you're going to query based on location.
 - Properties: `location_name`

2. Relationships:

- FRIENDS_WITH: Represents the friendship relationship between two users.
 - Properties: `since` (the date when the friendship started)
- CREATED: Represents a user creating a post.
 - Properties: `created_at` (timestamp when the post was created)
- COMMENTED: Represents a user commenting on a post.
 - Properties: `commented_at` (timestamp when the comment was made)
- LOCATED_IN: (Optional) Connects a user to a location.
 - Properties: `since` (timestamp when the user joined this location)

3. Queries:

- a. Retrieve all friends of a specific user.
- b. Find the most popular post (by the number of comments or likes).
- c. List all posts by a specific user and their associated comments.
- d. Identify the shortest path between two users (mutual friends or common connections).

- e. List all posts made by users who are located in a specific region.
- f. Find users who have commented on more than 5 posts in the last month.
- g. Find users who are the most connected (central users in the graph).
- h. Find mutual friends between two users.